

Modulus and Argument of a Complex Number



Mathematics | JEE Main & Advanced

Don't Memorize. Visualize.

A complex number is a point, and also an arrow from the origin. Its **modulus** is how long the arrow is. Its **argument** is which way the arrow points.

MODULUS

How long the arrow is. Also called the magnitude.

Written $|z|$.

ARGUMENT

Which way the arrow points. Also called the angle.

Written $\arg z$.

A complex number is a point and an arrow

The complex number $z = x + iy$ is the point (x, y) on the Argand plane: the real part x along the real axis (Re) and the imaginary part y along the imaginary axis (Im). Join the origin to the point and you get an arrow. For $z = 3 + 4i$ the arrow and the two sides 3 and 4 form a right triangle.

Modulus: how long the arrow is

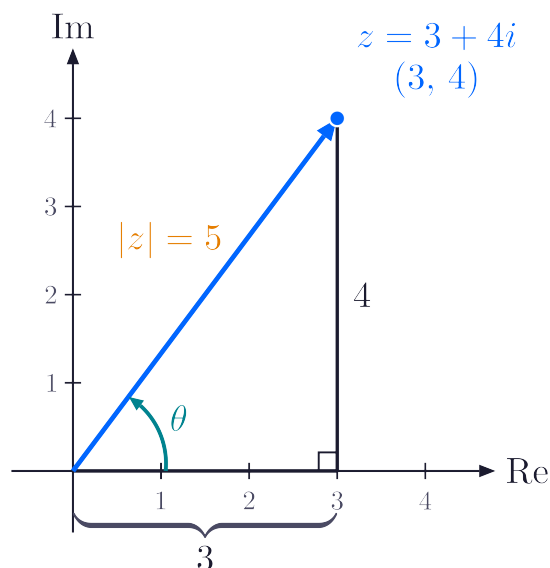
By Pythagoras,

$$|z| = \sqrt{x^2 + y^2}, \quad |3 + 4i| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$

It is a length, so it is never negative, and it is 0 only for $z = 0$.

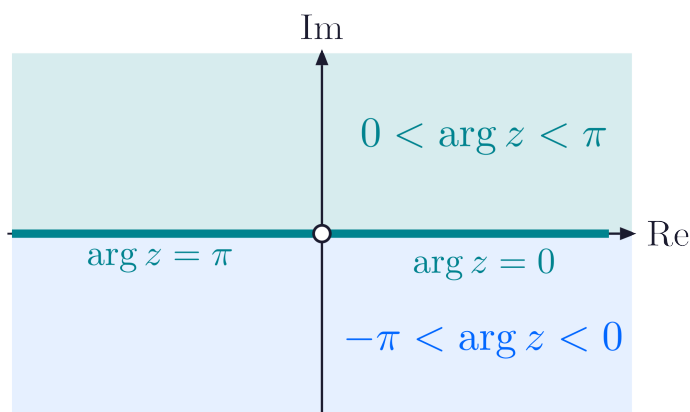
Argument: which way the arrow points

The angle $\arg z$ from the positive real axis to the arrow. Anticlockwise counts as positive, clockwise as negative. For $3 + 4i$, $\tan \theta = \frac{4}{3}$, so $\theta = \tan^{-1}\left(\frac{4}{3}\right) \approx 0.9273$ radians, about 53.13° . For $z = 0$ there is no arrow, so $\arg 0$ is not defined.

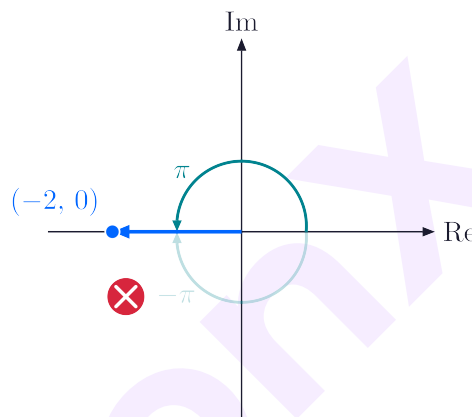


The principal argument

One arrow has many angles: turn the other way round, or add full turns of 2π . We agree to report one of them, the **principal argument**, in $(-\pi, \pi]$: the shorter turn wins.



Above the real axis the argument is between 0 and π , below it between $-\pi$ and 0. The origin has no argument.



On the negative real axis both turns are half a turn. The agreement keeps the anticlockwise one: $\arg(-2) = \pi$, never $-\pi$.

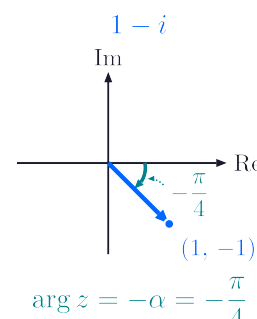
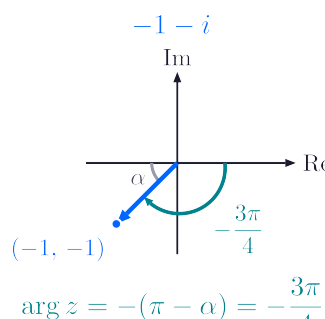
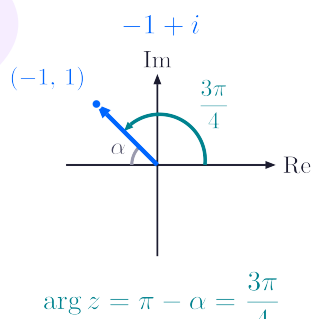
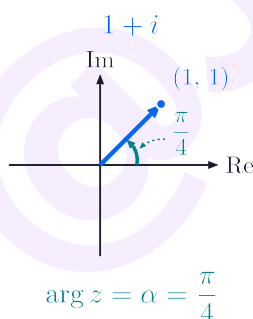
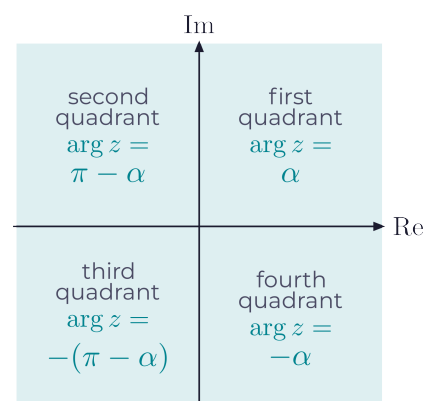
Finding the argument in any quadrant

STEP 1 Find the acute angle the arrow makes with the real axis,

$$\alpha = \tan^{-1} \left| \frac{y}{x} \right|.$$

STEP 2 Read the argument from the quadrant the point is in, with the chart.

A point on an axis is in no quadrant: read its argument from the picture. It is 0 on the positive real axis, π on the negative real axis, $\frac{\pi}{2}$ on the imaginary axis above the origin and $-\frac{\pi}{2}$ below it.



The same acute angle $\alpha = \frac{\pi}{4}$ in all four quadrants gives four different arguments.

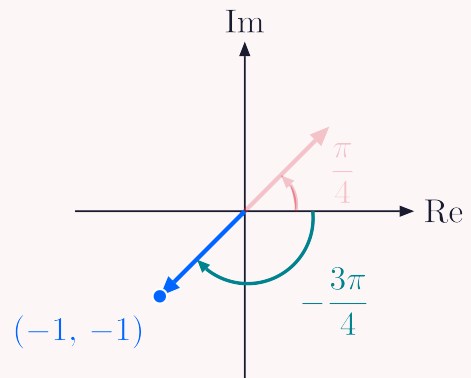
CAREFUL

For $z = -1 - i$ a calculator gives

$$\tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1}{-1}\right) = \tan^{-1}(1) = \frac{\pi}{4}.$$

But an arrow at $\frac{\pi}{4}$ points to $(1, 1)$ in the first quadrant, and $-1 - i$ is the point $(-1, -1)$ in the third. Dividing y by x threw away both signs.

The true argument is $-\left(\pi - \frac{\pi}{4}\right) = -\frac{3\pi}{4}$. Always check the quadrant.



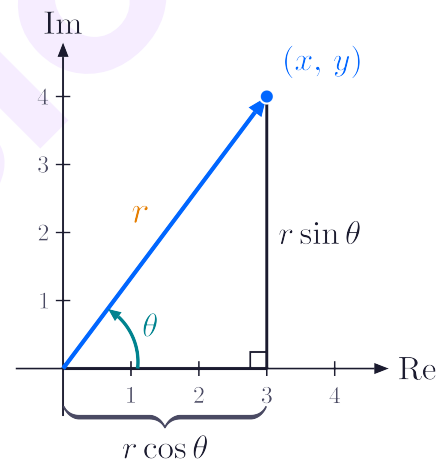
Polar form

The two legs of the triangle are $x = r \cos \theta$ and $y = r \sin \theta$, where $r = |z|$ and $\theta = \arg z$. Put them into $z = x + iy$:

Polar form

$$z = r(\cos \theta + i \sin \theta)$$

For our example, $3 + 4i = 5(\cos \theta + i \sin \theta)$ with $\theta = \tan^{-1}\left(\frac{4}{3}\right)$.



Solved example

Find $|z|$ and $\arg z$ for $z = -1 - \sqrt{3}i$.

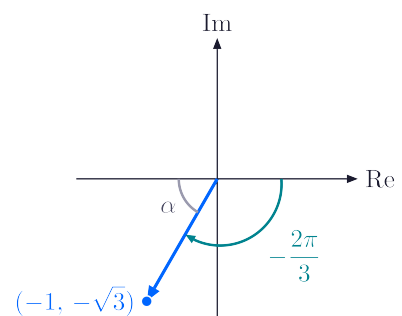
Plot it. The point is $(-1, -\sqrt{3})$, in the third quadrant.

Modulus. $|z| = \sqrt{(-1)^2 + (-\sqrt{3})^2} = \sqrt{1+3} = 2$.

Acute angle. $\alpha = \tan^{-1}\left|\frac{-\sqrt{3}}{-1}\right| = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$.

Argument. Third quadrant, so $\arg z = -\left(\pi - \frac{\pi}{3}\right) = -\frac{2\pi}{3}$.

$$|z| = 2, \quad \arg z = -\frac{2\pi}{3}$$



Key formulas

$$|z| = \sqrt{x^2 + y^2}$$

$$\alpha = \tan^{-1} \left| \frac{y}{x} \right|$$

first quadrant: $\arg z = \alpha$

second quadrant: $\arg z = \pi - \alpha$

third quadrant: $\arg z = -(\pi - \alpha)$

fourth quadrant: $\arg z = -\alpha$

principal argument: $-\pi < \arg z \leq \pi$

polar form: $z = r(\cos \theta + i \sin \theta)$

Practice set

Easy to hard. Give every argument as the principal argument, in radians. Sketch each point first: the picture tells you the quadrant. The answers start on the next page.

- Q1.** Find $|z|$ and $\arg z$ for $z = 1 + \sqrt{3}i$.
- Q2.** Find $|z|$ and $\arg z$ for $z = -3$.
- Q3.** Find $|z|$ and $\arg z$ for $z = -2i$.
- Q4.** Find $|z|$ and $\arg z$ for $z = \sqrt{3} - i$.
- Q5.** Find $|z|$ and $\arg z$ for $z = \frac{1+i}{1-i}$.
- Q6.** Write $z = -1 + \sqrt{3}i$ in polar form.
- Q7.** Find $|z|$ and $\arg z$ for $z = -4 - 4i$.
- Q8.** Let $z = -1 - i$. Find $\arg(z^2)$. Is it equal to $2 \arg z$?



Watch the animated lesson on YouTube

Modulus and Argument of a Complex Number

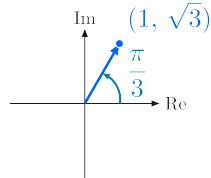
Scan the code to open the JEEVisionX channel on YouTube. This lesson is in the Complex Numbers playlist.

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Answers

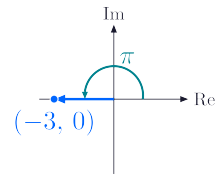
Q1

$|z| = 2, \arg z = \frac{\pi}{3}$
 $|z| = \sqrt{1+3} = 2$. First quadrant, so $\arg z = \alpha = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$.



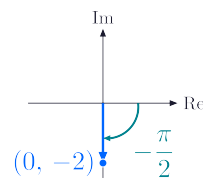
Q2

$|z| = 3, \arg z = \pi$
 The point $(-3, 0)$ is on the negative real axis. The argument is π , not $-\pi$.



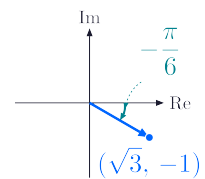
Q3

$|z| = 2, \arg z = -\frac{\pi}{2}$
 The point $(0, -2)$ is straight down the imaginary axis: a quarter turn clockwise.



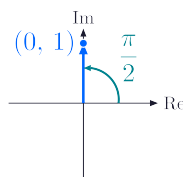
Q4

$|z| = 2, \arg z = -\frac{\pi}{6}$
 $|z| = \sqrt{3+1} = 2$. Fourth quadrant, $\alpha = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$, so $\arg z = -\frac{\pi}{6}$.



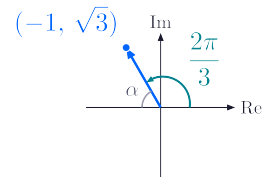
Q5

$|z| = 1, \arg z = \frac{\pi}{2}$
 Multiply top and bottom by $1+i$: $\frac{(1+i)^2}{(1-i)(1+i)} = \frac{2i}{2} = i$, the point $(0, 1)$.



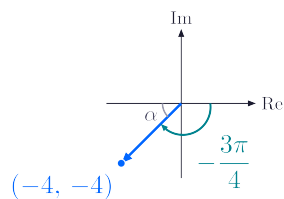
Q6

$z = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$
 $r = \sqrt{1+3} = 2$. Second quadrant, $\alpha = \frac{\pi}{3}$, so $\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$.



Q7

$|z| = 4\sqrt{2}, \arg z = -\frac{3\pi}{4}$
 $|z| = \sqrt{16+16} = 4\sqrt{2}$. Third quadrant, $\alpha = \frac{\pi}{4}$, so $\arg z = -(\pi - \frac{\pi}{4}) = -\frac{3\pi}{4}$, not $\frac{\pi}{4}$.



Q8

$\arg(z^2) = \frac{\pi}{2}$, and no.
 $z^2 = (-1-i)^2 = 1+2i+i^2 = 2i$, so $\arg(z^2) = \frac{\pi}{2}$. But $2 \arg z = 2(-\frac{3\pi}{4}) = -\frac{3\pi}{2}$ (the grey turn), outside $(-\pi, \pi]$. Same direction, a full turn of 2π apart, but only $\frac{\pi}{2}$ is the principal argument.

